FMEA

Process/Product Name: ______________________ Prepared By: ______________________

Responsible: ______________________ FMEA Date (Orig.): ____________ (Rev.): ____________

Process Step/Input	Potential Failure Mode	Potential Failure Effects	SEVERITY (1 - 10)	Potential Causes	OCCURRENCE (1 - 10)	Current Controls	DETECTION (1 - 10)	RPN	Action Recommended	Resp.	Actions Taken	SEVERITY (1 - 10)	OCCURRENCE (1 - 10)	DETECTION (1 - 10)	RPN
What is the process step, change or feature under investigation?	In what ways could the step, change or feature go wrong?	What is the impact on the customer if this failure is not prevented or corrected?		What causes the step, change or feature to go wrong? (how could it occur?)		What controls exist that either prevent or detect the failure?			What are the recommended actions for reducing the occurrence of the cause or improving detection?	Who is responsible for making sure the actions are completed?	What actions were completed (and when) with respect to the RPN?				
ATM PIN Authentification	*Unauthorized access*	*unauthorized cash withdrawal & very*						*0*							*0*
	Authentification failure							0							0
Dispense cash	cash not disbursed							0							0
	account debited but cash not disbursed							0							0
	extra cash dispensed							0							0
								0							0

FMEA

Process/Product Name: Coffee Process

Prepared By: Tracy

Responsible: Servers

FMEA Date (Orig.): May 15th (Rev.): May 30th

Process Step/Input	Potential Failure Mode	Potential Failure Effects	SEVERITY (1 - 10)	Potential Causes	OCCURRENCE (1 - 10)	Current Controls	DETECTION (1 - 10)	RPN	Action Recommended	Resp.	Actions Taken	SEVERITY (1 - 10)	OCCURRENCE (1 - 10)	DETECTION (1 - 10)
What is the process step or feature under investigation?	In what ways could the step or feature go wrong?	What is the impact on the customer if this failure is not prevented or corrected?		What causes the step or feature to go wrong? (how could it occur?)		What controls exist that either prevent or detect the failure?			What are the recommended actions for reducing the occurrence of the cause or improving detection?	Who is responsible for making sure the actions are completed?	What actions were completed (and when) with respect to the RPN?			
Fill carafe with water	Wrong amount of water	Coffee too strong or weak	8	Faded level marks on carafe	4	Visual Inspection	4	128	Replace old carafes	Mel	Carafe replaced 9/15	8	1	3
			8	Water spilled from carafe	5	None	9	360	Train employees	Flo	Trained on 9/20	8	2	7
	Water too warm	Coffee too strong	8	Faucet not allowed to run cool	8	Finger	4	256	Train employees	Flo	Trained on 9/20	8	2	6
			8	Employee unaware of need for cool water	7	None	7	392	Train employees	Flo	Trained on 9/20	8	1	8
	Unclean carafe	Foreign objects in coffee	10	Carafe not washed	4	Visual Inspection	4	160	Appoint inspector before storage	Alice	Vera appointed 9/21	10	1	4
		Bad tasting coffee	10	Carafe stored improperly	7	Training	5	350	Create storage & instructions	Alice	Bin & visual instructions done 9/25	10	2	3

FMEA Form

FMEA Form

v3.6

RPN
24
112
96
64
40
60

FMEA Form

FMEA

Process/Product Name: Food Order Delivery
Prepared By: Team Lead
Responsible: Manager
FMEA Date (Orig.): May 1st (Rev.): May 5th

Process Step/Input	Potential Failure Mode	Potential Failure Effects	SEVERITY (1 - 10)	Potential Causes	OCCURRENCE (1 - 10)	Current Controls	DETECTION (1 - 10)	RPN	Action Recommended	Resp.	Actions Taken	SEVERITY (1 - 10)	OCCURRENCE (1 - 10)	DETECTION (1 - 10)
What is the process step or feature under investigation?	In what ways could the step or feature go wrong?	What is the impact on the customer if this failure is not prevented or corrected?		What causes the step or feature to go wrong? (how could it occur?)		What controls exist that either prevent or detect the failure?			What are the recommended actions for reducing the occurrence of the cause or improving detection?	Who is responsible for making sure the actions are completed?	What actions were completed (and when) with respect to the RPN?			
Changing Vendors	Run out of key food products	Unable to offer standard menu items	8	Supply chain issues	8	Won't know until delivery	10	640	Give vendor a 30-day trial	Manager	Directions created, copied and distributed on May 5th	8	6	10
			8	Delivering to others first	8	Won't know until delivery	10	640	Have vendor call to confirm quantities	Manager	Instituted on May 5th	8	4	10
			8		8		10	640	Impose penalties for missed items	Manager	Instituted on May 6	8	4	10
	Products may differ from what's currently served	Customers reject the new products	10	The vendor has found cheaper sources	9	Won't know until customers taste new product	10	900	Conduct taste tests on any new brands of products - reject as needed	Chef	Setup testing trials on May 5th	10	4	10

			10	Vendor delivers best food to long-term customers	7	Won't know until customers taste new product	10	700	Have vendor replace any products that don't meet requirements	Manager	Contact vendor as needed on May 6	8	4	10
								0						
								0						
								0						
								0						
								0						

FMEA Form

v3.6

RPN
480
320
320
400

320
0
0
0
0
0

Severity Scale

Adapt as appropriate

Effect	Criteria: Severity of Effect	Ranking
Hazardous - Without Warning	May expose client to loss, harm or major disruption - failure will occur **without** warning	10
Hazardous - With Warning	May expose client to loss, harm or major disruption - failure will occur **with** warning	9
Very High	Major disruption of service involving client interaction, resulting in either associate re-work or inconvenience to client	8
High	Minor disruption of service involving client interaction and resulting in either associate re-work or inconvenience to clients	7
Moderate	Major disruption of service not involving client interaction and resulting in either associate re-work or inconvenience to clients	6
Low	Minor disruption of service not involving client interaction and resulting in either associate re-work or inconvenience to clients	5
Very Low	Minor disruption of service involving client interaction that does not result in either associate re-work or inconvenience to clients	4
Minor	Minor disruption of service not involving client interaction and does not result in either associate re-work or inconvenience to clients	3
Very Minor	No disruption of service noticed by the client in any capacity and does not result in either associate re-work or inconvenience to clients	2
None	No Effect	1

Occurrence Scale

Probability of Failure	Time Period	Per Item Failure Rates	Ranking
Very High: Failure is almost inevitable	More than once per day	>= 1 in 2	10
	Once every 3-4 days	1 in 3	9
High: Generally associated with processes similar to previous processes that have often failed	Once every week	1 in 8	8
	Once every month	1in 20	7
Moderate: Generally associated with processes similar to previous processes which have experienced occasional failures, but not in major proportions	Once every 3 months	1 in 80	6
	Once every 6 months	1 in 400	5
	Once a year	1 in 800	4
Low: Isolated failures associated with similar processes	Once every 1 - 3 years	1 in 1,500	3
Very Low: Only isolated failures associated with almost identical processes	Once every 3 - 6 years	1 in 3,000	2
Remote: Failure is unlikely. No failures associated with almost identical processes	Once Every 7+ Years	1 in 6000	1

Detection Scale

Detection	Criteria: Likelihood the existence of a defect will be detected by process controls before next or subsequent process, -OR- before exposure to a client	Ranking
Almost Impossible	No known controls available to detect failure mode	10
Very Remote	Very remote likelihood current controls will detect failure mode	9
Remote	Remote likelihood current controls will detect failure mode	8
Very Low	Very low likelihood current controls will detect failure mode	7
Low	Low likelihood current controls will detect failure mode	6
Moderate	Moderate likelihood current controls will detect failure mode	5
Moderately High	Moderately high likelihood current controls will detect failure mode	4
High	High likelihood current controls will detect failure mode	3
Very High	Very high likelihood current controls will detect failure mode	2
Almost Certain	Current controls almost certain to detect the failure mode. Reliable detection controls are known with similar processes.	1